Dr. Daniel Thomas

Akademischer rat a.Z.
Lehrstuhl f. Ökologische
Mikrobiologie
Universität Bayreuth

Telefon +49(0)921/55-5620
e-mail: daniel.thomas@uni-bayreuth.de
https://www.bayceer.uni-bayreuth.de/bayceer/

Dr. Hans-Frisch-Straße 1-3
95448 Bayreuth,
Germany

To the Editorial team of Oecologia,

Please find attached our manuscript, “Scaling up by looking closer: extending the range of moss-Nostoc symbioses deeper into central Europe and to new moss hosts in the forest canopy.”

In this project we searched for – and found – the well-known feather-moss/*Nostoc* symbioses in our region of Germany. As far as we are aware, we are the first to provide detailed observations of this symbiosis in our region and in Germany, complete with microscopic evidence and genetically identified cultures of the microoorganisms involved. The likely presence of *Nostoc* in forest mosses has been incidentally noted in one environmental metabarcode study (see our introduction), but was not further confirmed.

Regardless of the possible geographical extension of this symbioses, we also make several other interesting observations. We used Nanopore sequencing to observe the unexpected presence of multiple possible nifH genes within our *Nostoc* cultures, which causes us to call for the re-interpretation of the results of important previous ecological studies on moss-*Nostoc* nifH studies by other groups. We contextualize the feather-moss/*Nostoc* symbioses into the increasing awareness that all mosses generally host a rich community of cyanobacteria. We present PCR-based evidence for the possible presence of cyanobacterial nitrogenases throughout the aerial tissues of a feather mosses, despite lack of visible colonization by *Nostoc*, and use epifluorescent microscopic to provide a possible explanation for this: possible cyanobacterial endophytes in the “stems” (caulids) of our mosses.

We hope Oecologia finds this manuscript of sufficient interest for its readers. We look forward to your response.

Sincerely,

Daniel Thomas
19 January 2026